

Kevin Tongue

DOUBLE DEGREE – MECHANICAL ENGINEERING & MASTER IN NUMERICAL SOLID MECHANICS

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Education

 **Centrale Lyon – ENISE** &  **École Nationale Supérieure Polytechnique de Yaoundé** *Saint-Étienne, France / Yaoundé, Cameroon*
(ENSPY)

DOUBLE DEGREE – ENGINEERING IN MECHANICAL ENGINEERING & MASTER 2 IN NUMERICAL SOLID MECHANICS

2020 – 2026 (expected)

- **Specialisation:** Mechanical design, advanced numerical simulation, and computational mechanics.
- **Key courses (Master):** Finite Element Method, Continuum Mechanics, Fracture Mechanics, Structural Dynamics, Nonlinear Mechanics, Uncertainty Quantification, Design of Experiments, Residual Stress Analysis, Fluid Dynamics, Heat Transfer, Materials Science.
- **Key courses (Engineering):** Solid Mechanics, Thermodynamics, Machine Design, Manufacturing Processes, System Dynamics, Signal Processing, Applied Mathematics.

Research & Professional Experience

 **IMT Mines Albi** &  **IMT Mines Saint-Étienne** *Albi / Saint-Étienne, France*

MASTER INTERNSHIP – DETERMINISTIC-STOCHASTIC COUPLING FOR GRANULAR SEGREGATION (DEM + MARKOV CHAINS)

Mar. 2026 – Sep. 2026

- Coupled Discrete Element Method (DEM) simulations with an inhomogeneous Markov chain model to statistically characterise granular segregation dynamics in mixing devices.
- Extracted particle-scale transition probabilities from DEM outputs; developed stochastic models linking microscopic particle dynamics to macroscopic distributions.
- Designed and executed a Design of Experiments (DoE) plan and performed Monte Carlo sensitivity analysis for uncertainty quantification (UQ).
- Implemented end-to-end numerical pipelines in Python (NumPy, SciPy, PyTorch, Matplotlib) including data fusion, signal processing, and model validation.

 **Centrale Lyon – ENISE** *Saint-Étienne, France*

RESEARCH PROJECT – NONLINEAR FEM SOLVER FOR FATIGUE FAILURE PREDICTION (DANG VAN CRITERION)

Oct. 2025 – Mar. 2026

- Implemented a 3D nonlinear finite element solver in Python/C++ covering static, modal, and transient dynamic analyses.
- Applied and validated the Dang Van multiaxial fatigue criterion, including critical plane determination (Agard et al. optimisation methodology).
- Validated results against analytical benchmarks and Abaqus reference solutions; produced structured technical documentation.
- Deployed as a web application: huggingface.co/spaces/ktongue/simulations_apps

 **Centrale Lyon – ENISE (collaboration with LTDS Laboratory)** *Saint-Étienne, France*

RESEARCH PROJECT – PHYSICS-INFORMED NEURAL NETWORKS (PINN) FOR MATERIAL IDENTIFICATION

2025

- Built a Physics-Informed Neural Network (PINN) in PyTorch coupling deep learning with physical constraints (elasticity, heat equation, plasticity criteria).
- Performed inverse identification of constitutive law parameters from compression test data provided by the LTDS laboratory; validated the identified law via FEM simulations.
- Implemented data-driven surrogate modelling pipeline: preprocessing, training loop, hyperparameter tuning, and result visualisation.
- Deployed at: huggingface.co/spaces/ktongue/material_identification

 **Centrale Lyon – ENISE** *Saint-Étienne, France*

LABORATORY COURSEWORK – SURFACE INTEGRITY AND RESIDUAL STRESS CHARACTERISATION IN MACHINING

2023 – 2024

- Measured residual stresses induced by machining operations using laser-based spectroscopy techniques, and correlated results with cutting parameters and tool wear state.
- Conducted experimental micro-hardness mapping (Vickers micro-indentation) on machined 42CrMo4 steel specimens, comparing surface and core hardness across new vs. worn cutting inserts and parallel vs. perpendicular cutting directions.
- Characterised microstructure and tool wear mechanisms using Scanning Electron Microscopy (SEM) and Energy-Dispersive X-ray Spectroscopy (EDXS) to identify coating degradation and material composition changes at cutting edges.
- Analysed the influence of machining operations on surface integrity (hardness gradients, microstructural alteration, residual stress state) to define recommendations ensuring part integrity.

 **École des Mines de Saint-Étienne** *Saint-Étienne, France*

ACADEMIC PROJECT – MICROSTRUCTURE PARTITIONING MODEL (IMAGE SEGMENTATION)

Feb. – Jun. 2025

- Processed and segmented microstructural images using Python (NumPy, OpenCV, SciPy); performed geometric feature extraction and statistical characterisation of material phases.
- Developed modular segmentation and classification algorithms applicable to materials with heterogeneous microstructures.



Saint-Étienne, France

- Pre-dimensioning study, strength of materials calculations, and 3D CAD modelling (SolidWorks); ran static finite element simulation (SolidWorks Simulation) to validate structural integrity.
- Produced full technical documentation including functional analysis, tolerancing, and manufacturing drawings.



Saint-Étienne, France

- Designed and developed a full-stack library management web application using Symfony (PHP); implemented MVC architecture, database design, and user authentication.



Saint-Étienne, France

- Mechanical design and CAD modelling with SolidWorks and CATIA; functional analysis, ergonomic study, component selection, and technical documentation.

Yaoundé, Cameroon

May 2024 – Sep. 2024

- Investigated root causes of pump failures on KBA newspaper printing machines; proposed corrective and preventive maintenance strategies.
- Designed a software package for managing the preventive maintenance schedule of printing press pumps.
- Instrumented machines with vibration and temperature sensors; performed signal processing and failure mode analysis (FMEA approach).



Yaoundé, Cameroon

May – Sep. 2023

- Full CAD design (SolidWorks: assemblies, sheet metal, machining preparation) and manufacturing of a 3-axis CNC machine for prototyping and educational use.
- Structural sizing, component selection (linear guides, stepper motors, belt transmission), and assembly supervision.

Technical Skills

Continuum mechanics, fracture mechanics, fatigue (Dang Van, Goodman), thermo-elasticity, cohesive zone models (Dugdale, DDM, CZM), residual stress analysis (laser spectroscopy), constitutive law identification, multi-scale modelling

Vickers micro-indentation hardness testing, Scanning Electron Microscopy (SEM), Energy-Dispersive X-ray microstructure analysis, surface integrity assessment in machining.

Abaqus (static, dynamic, thermal), SolidWorks Simulation, ANSYS Workbench (basics), Cast3M (basics), commercial solver (Python/C++): static, modal, and transient analyses.

Discrete Element Method (DEM/LIGGGHTS), Markov chain modelling, Monte Carlo simulation, Design of Experiments, uncertainty quantification, surrogate modelling.

Physics-Informed Neural Networks (PINNs, PyTorch), inverse parameter identification, data-driven microscale modeling, and deep learning for scientific computing.

Python (NumPy, SciPy, PyTorch, Matplotlib, OpenCV, Pandas), C/C++, MATLAB (Simulink, Computer Vision
toolboxes), Fortran (basics), Java, LaTeX, Git/GitHub.

SolidWorks (assemblies, sheet metal, simulation), CATIA V5 (part design, surfacing, PLM), 3DExperience p

Sensors (accelerometers, thermocouples, IMU, encoders), data acquisition, vibration analysis, GR&R, Cp/analysis.

Python-based web applications (Gradio, Hugging Face Spaces), Symfony/PHP (MVC web apps), basic HTML

French (native), English (B2 – professional technical proficiency, TOEIC), German (beginner).

Certifications & Activities



Saint-Étienne, France

- Integration of sensors (IMU, encoders, temperature sensors), real-time data processing, firmware development, and collaborative project management.

MathWorks

Google / Coursera

IBM / Coursera

Open-Source & Deployed Projects

2026 **FEM Fatigue Solver**, Nonlinear 3D FEM solver with Dang Van fatigue criterion – Python/C++ backend, interactive web interface

huggingface.co/spaces/karimkhanlou/fem-fatigue-solver

2025 **PINN Material Identification**, Physics-Informed Neural Network for inverse identification of constitutive law parameters from compression tests

huggingface.co/spaces/karimkhanlou/pinn-material-identification

Additional Information

- **Availability:** September/October 2026 – full geographic mobility (France and internationally).
- **Research interests:** Computational mechanics, granular media, fracture and fatigue, inverse problems, scientific machine learning, non-destructive evaluation (NDE), multi-physics simulation.
- **Soft skills:** Scientific rigour, analytical mindset, research autonomy, fast learner, team player, problem-solving orientation.